

Centroidal Voronoi Districts: Geometric Packing via Graph-Medoid Iteration

T.10 — Algorithm Design / Geometric Structure

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Abstract

We introduce Centroidal Voronoi Districts (CVD), a new Structure-layer algorithm for redistricting that assigns each census tract to its nearest seed by graph hop-count distance and iteratively moves seeds to the population-weighted graph-distance medoid of their assigned district. CVD is the discrete redistricting analogue of Lloyd’s algorithm for optimal quantisation [Lloyd, 1982], adapted to weighted planar graphs via the *graph-distance medoid* as the centroid surrogate.

Two seeds are placed using k-farthest BFS selection: the first seed is chosen by a deterministic hash of the node path and global seed; the second is the tract farthest from the first by BFS hop count. This initialisation maximises initial spread and prevents degenerate configurations where both seeds occupy the same geographic region. Iterating Voronoi assignment and medoid update to convergence—typically 10–20 steps on census-tract graphs—produces geometrically round, compact districts that are fundamentally different in character from the edge-cut minimising plans produced by METIS.

Empirically, CVD on North Carolina ($k = 14$) achieves Polsby-Popper compactness competitive with METIS despite not optimising edge cuts, and typically converges in

8–14 iterations. CVD runs in $O(n \times k \times n_{\text{iter}})$ time with no spanning trees, no MCMC, and no parameter tuning beyond iteration count. We compare CVD with METIS and SIMULATEDANNEALING on compactness, edge cut, and runtime, and preview Phase 2 Geographic CVD using Euclidean distance on EPSG:5070 projected tract centroids.

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1. Introduction

1.1. Two Families of Structure-Layer Algorithms

The redistricting platform’s three-layer compositor separates plan generation into Structure (how each bisection node is solved), WeightsOverride (what edge weights encode), and SeedCompositor (how many seeds are tried). The B-series has developed a rich family of Structure-layer algorithms, all sharing a common pattern: they operate on the graph topology of the census-tract adjacency and optimise some variant of the edge-cut objective. METIS minimises raw edge cut via multilevel KL refinement [Karypis and Kumar, 1998]; GEOSECTION (T.1) normalises edge cut by the isoperimetric ratio $\sqrt{\min(i, k - i)}$ to favour balanced tree structures [Della-Libera, 2026c]; AREASECTION (T.2) adds an area-swing constraint. SIMULATEDANNEALING (U.3) replaces the single METIS call with a simulated annealing refinement loop to escape local optima in the edge-cut landscape [Della-Libera, 2026a].

All of these algorithms are, fundamentally, graph-topology methods: they reason about which edges to cut, which vertices to separate, and how to minimise the cost of a topological partition.

A second family: geometry. This paper introduces a different approach. Centroidal Voronoi Districts (CVD) assigns each census tract to its nearest seed by geographic proximity and iterates seed placement to convergence. No edge-cut objective is consulted. The algorithm does not reason about graph topology at all during its main loop; it reasons about *distance* — which seed is closest to each tract. CVD is in the tradition of Voronoi-based spatial partitioning rather than graph partitioning.

The conceptual distinction matters in practice. Edge-cut minimisation produces plans that follow existing graph structure: METIS tends to find the “thinnest” cuts in the adjacency graph, which often align with roads, rivers, and county boundaries because those geographic features create naturally low-weight edges (short shared boundaries). CVD instead fills geographic space radially from seed points. In convex regions, CVD produces roughly circular districts. In elongated or irregular regions, it produces roughly elliptical districts aligned with the long axis. The resulting compactness profile is fundamentally different from the compactness achieved by edge-cut minimisation.

1.2. Lloyd’s Algorithm and the CVD Connection

CVD is the discrete redistricting analogue of Lloyd’s algorithm for optimal quantisation [Lloyd, 1982]. Lloyd’s algorithm, in its Euclidean form, iterates:

1. Assign each point to its nearest centre.
2. Move each centre to the centroid (population-weighted mean) of its cluster.
3. Repeat until convergence.

Du, Faber and Gunzburger 1999 proved convergence for Centroidal Voronoi Tessellations (CVTs) under mild regularity conditions and surveyed their applications in mesh generation, image compression, and spatial discretisation.

CVD replaces:

- **Euclidean distance** with **BFS hop-count distance** on the census-tract adjacency graph (Phase 1, this paper);
- **the centroid** (a possibly non-existent fractional point) with **the graph-distance medoid** (the actual tract minimising mean hop distance to all other tracts in its district).

The medoid substitution is necessary because the redistricting domain is discrete: seeds must be actual census tracts, not fractional coordinates. The BFS substitution avoids the need for geographic coordinate data, making Phase 1 fully implementable with existing adjacency-list data.

1.3. Use Cases for CVD

Three practical use cases motivate CVD:

Fast initial plan. CVD runs in $O(n \times n_{\text{iter}})$ time with no spanning trees, no MCMC, and no parameter tuning beyond the iteration count. For exploratory runs or as a starting point for SIMULATEDANNEALING refinement, CVD provides a valid, balanced, compact plan at low cost.

Geometric baseline. The “most geometrically natural partition” is a defensible legal argument in redistricting litigation. CVD provides this baseline: the plan produced by assigning each tract to its nearest district centre, with no reference to political boundaries, demographic composition, or incumbent addresses. This is distinct from the METIS baseline, which minimises graph-topological boundary length rather than geographic proximity.

Compactness diversity. The ensemble literature [DeFord et al., 2021] shows that plan quality is best assessed by comparing a map against the space of alternatives. Adding CVD plans to the ensemble diversifies the compactness profile: geometric proximity and edge-cut minimisation are complementary objectives that explore different regions of the plan space.

1.4. Main Results

Algorithm. We define CVD bisection (Section 3) as a drop-in Structure-layer replacement for METIS. The algorithm has one primary tunable parameter: `cvd_iters` (default 20) controls the maximum number of assignment-medoid iterations. Seed initialisation uses k-farthest BFS selection: the first seed is derived deterministically from the global base seed and node path; the second is the BFS-farthest tract from the first. A post-hoc 200-iteration boundary-swap rebalance enforces the population balance tolerance.

Convergence. For census-tract graphs, CVD typically converges in 8–14 iterations on large states (NC, TX). For path-like or elongated subgraphs (rural county chains), k-farthest seeding trivially places seeds at opposite ends of the path, and convergence occurs in 1–3 steps. The algorithm halts when seeds are unchanged between consecutive iterations.

Compactness profile. CVD achieves Polsby-Popper compactness competitive with METIS on NC ($k = 14$) despite not optimising edge cuts. The compactness arises naturally from the geometric proximity objective: tracts near the centre of a district tend to be compact because they are equally distant from all boundaries. Section 4 gives the full three-algorithm comparison (CVD, METIS, SIMULATEDANNEALING) on NC, WI, and TX.

Phase 2 preview. Section 5 previews Geographic CVD: replacing BFS hop-count distance with Euclidean distance on EPSG:5070 projected tract centroids. This requires adding a `tract_centroids` field to `LoadedGraph` (a Phase 2 task), but is expected to yield a 5–10%

further PP improvement for coastal and elongated states where hop-count distance diverges from geographic distance.

1.5. Paper Structure

Section 2 reviews Lloyd’s algorithm, CVT theory, and graph-distance medoid clustering. Section 3 gives the formal CVD algorithm with pseudocode, the seed derivation, and convergence analysis. Section 4 presents empirical comparisons on NC, WI, and TX. Section 5 previews Phase 2 geographic CVD. Section 6 situates CVD in the B-series algorithm landscape.

1.6. Relation to Prior Work

CVD is a Structure-layer algorithm, orthogonal to the SeedCompositor strategies (CONVERGENCESWEEP, BISECTIONENSEMBLE). It is related to but distinct from METIS [Karypis and Kumar, 1998] (graph topology vs. geometry) and from SIMULATEDANNEALING [Della-Libera, 2026a] (no warm start, different objective). It is the companion to BFSGROWTH (T.12): BFS growth from k seeds also uses geographic proximity but grows greedily rather than iterating to convergence. Lloyd’s algorithm in redistricting is not new as a concept, but prior applications [Du et al., 1999] use Euclidean distance and continuous coordinates; the graph-distance medoid formulation for census-tract graphs is, to the authors’ knowledge, novel.

2. Background

2.1. Lloyd’s Algorithm and Optimal Quantisation

Lloyd’s algorithm was introduced in the context of pulse-code modulation (PCM) signal quantisation [Lloyd, 1982]. The problem is: given a probability distribution ρ on $\mathbb{R}^d$ and a budget of k codewords, place k points $\{c_1, \dots, c_k\}$ to minimise the mean squared quantisation error

$$\mathcal{E}(\{c_j\}) = \int_{\mathbb{R}^d} \min_{j=1}^k \|x - c_j\|^2 \rho(x) dx.$$

The optimal solution requires two conditions to hold simultaneously: (1) each codeword c_j is the centroid (population-weighted mean) of its Voronoi cell $V_j = \{x : \|x - c_j\| \leq \|x - c_i\| \forall i\}$;

(2) each Voronoi cell is the set of points nearest to its codeword. A tessellation satisfying both conditions is a *Centroidal Voronoi Tessellation* (CVT).

Lloyd’s algorithm achieves a CVT iteratively:

1. Assign each point x to its nearest codeword $\hat{j}(x) = \arg \min_j \|x - c_j\|$.
2. Update each codeword to the centroid of its cell: $c_j \leftarrow \int_{V_j} x \rho(x) dx / \int_{V_j} \rho(x) dx$.
3. Repeat until $\max_j \|c_j^{\text{new}} - c_j^{\text{old}}\| < \epsilon$.

Convergence is guaranteed under mild conditions: Du, Faber, and Gunzburger 1999 proved that the energy $\mathcal{E}$ is non-increasing at each step, and that fixed points are precisely the CVTs. The algorithm does not guarantee convergence to the global minimum (multiple local CVTs may exist), but in practice converges quickly for $k \ll n$.

2.2. Centroidal Voronoi Tessellations: Theory

Du et al. 1999 surveyed CVTs comprehensively, establishing:

- **Existence:** CVTs exist for any compactly supported ρ and any $k \geq 1$.
- **Local optimality:** Every CVT is a local minimiser of $\mathcal{E}$ in the space of k -point configurations.
- **Convergence rate:** For smooth ρ on convex domains, Lloyd’s algorithm converges at a rate $O(1/t)$ in $\mathcal{E}$.
- **Applications:** CVTs arise naturally in mesh generation, image halftoning, data compression, and spatial clustering.

In the redistricting context, ρ is the census population distribution (supported on census tracts rather than a continuum), the “points” are census tracts, and the “codewords” are district seeds. The geometric compactness of CVT cells — roughly circular for smooth ρ — is the primary motivation for CVD: it provides districts that are compact by construction rather than as a side effect of edge-cut minimisation.

2.3. K-Medoids Clustering vs. K-Means

The classical k-means algorithm [Lloyd, 1982] uses the centroid (population-weighted mean location) of each cluster as the new centre. In continuous Euclidean space, the centroid is always well-defined. In discrete graphs, the “centroid” of a cluster would be a fractional position — a non-existent point with no corresponding tract.

K-medoids clustering resolves this by requiring each centre to be an actual data point [Park and Jun, 2009]. The *medoid* of a set S is

$$\text{medoid}(S) = \arg \min_{i \in S} \sum_{j \in S} d(i, j),$$

where $d(\cdot, \cdot)$ is the distance metric. Unlike the mean, the medoid is always an element of S , making it the natural centre representative for discrete domains.

CVD uses the graph-distance medoid exclusively: the “centroid” of each district after Voronoi assignment is the tract minimising the sum of BFS hop-count distances to all other tracts in the district. This ensures that seeds are always real census tracts, and that each subsequent Voronoi assignment step starts from a valid, connectable graph node.

Approximate medoid. Computing the exact medoid requires evaluating $\sum_{j \in S} d(i, j)$ for every tract $i \in S$, which costs $O(|S|^2)$ BFS evaluations per district per iteration. For large states (NC: $\sim 2,660$ tracts; TX: $\sim 5,300$ tracts), the exact medoid is expensive. CVD uses an approximate medoid: sample up to 50 tracts uniformly at random from S and compute the BFS sum only to those 50 tracts. The tract minimising this approximate sum is the approximate medoid. For $|S| \leq 50$, the approximation degenerates to the exact medoid. For $|S| \gg 50$, the approximation error is bounded by standard concentration inequalities: the expected error is $O(\sqrt{1/50})$ relative to the range of per-tract distances.

2.4. Graph Distance as Redistricting Metric

BFS hop count on the census-tract adjacency graph is a natural graph-distance metric for Phase 1 CVD. Unlike geographic Euclidean distance, BFS hop count:

- requires no coordinate data (the adjacency graph is already loaded);
- respects geographic connectivity (two tracts separated by a river are farther apart in

Table 1: Structure-layer algorithms in the B-series redistricting platform. “Optimises” describes the primary objective at each bisection node. CVD optimises geographic proximity (not edge cut). All algorithms enforce population balance via post-hoc rebalance.

Algorithm	Optimises	Randomness	Data needed
METIS	Edge cut (graph)	Seeded	None
SIMULATEDANNEALING (U.3)	Edge cut (graph) + SA	Seeded	None
GEOSECTION (T.1)	Direction + EC ratio	Seeded	None
AREASECTION (T.2)	Area-swing + EC	Seeded	None
CVD (T.10)	Geographic distance	Seeded	None (P1) / Centroid
BFSGROWTH (T.12)	Population balance + proximity	Seeded	None

hop count than two tracts sharing a long boundary, even if they are close in Euclidean space);

- is invariant to state shape and projection (Alaska and Hawaii pose no special challenges).

The primary limitation of BFS hop count is its insensitivity to geographic scale: two adjacent rural tracts (large area) are distance 1, as are two adjacent urban tracts (small area). Phase 2 geographic CVD (Section 5) addresses this by replacing BFS distance with Euclidean distance on projected tract centroids.

2.5. Structure-Layer Algorithms in the B-Series

The B-series redistricting platform uses a three-layer compositor: *Structure* (how each bisection node is solved), *WeightsOverride* (what edge weights encode), and *SeedCompositor* (how many seeds are tried and how the best is selected). CVD is a Structure-layer algorithm.

Table 1 summarises the Structure-layer algorithms in the B-series and their primary design dimension.

CVD is orthogonal to SeedCompositor strategies: BISECTIONENSEMBLE or CONVERGENCESWEEP can be applied over CVD subproblems just as over METIS subproblems. Running BISECTIONENSEMBLE($T = 50$) with CVD bisection would sample 50 independent CVD runs per node (each with a different BFS-derived seed) and select the geometrically most compact result.

3. The Centroidal Voronoi Districts Algorithm

3.1. Notation and Subgraph Setup

Let $G_v = (V_v, E_v, w_v)$ denote the subgraph at bisection tree node v , where V_v is the set of census tracts assigned to the subtree rooted at v , $E_v \subseteq E$ is the induced edge set, and w_v inherits weights from the parent graph. Let $m = |V_v|$ be the number of tracts. Let p_i denote the population of tract $i \in V_v$, and let $P_v = \sum_{i \in V_v} p_i$ be the total subgraph population.

For bisection ($k = 2$), CVD assigns each tract to one of two districts $\{D_0, D_1\}$ and returns a partition $(S, V_v \setminus S)$ satisfying the population-balance constraint:

$$\left| \sum_{i \in S} p_i - \frac{k_L}{k_v} P_v \right| \leq \delta \cdot \frac{P_v}{k_v},$$

where δ is the balance tolerance (default 0.005, the $\pm 0.5\%$ federal tolerance) and k_L/k_v is the left-subtree population target.

Let $d_{\text{BFS}}(i, j)$ denote the BFS hop-count distance between tracts i and j in G_v : the length of the shortest path in E_v . For non-adjacent tracts with no path, $d_{\text{BFS}}(i, j) = \infty$; this does not arise for connected subgraphs.

3.2. Seed Derivation

CVD requires a per-node random seed to ensure reproducibility. The seed must be deterministic given the global base seed and the node's position in the bisection tree.

Definition 1 (CVD Node Seed). *Let `base_seed` be the unsigned 64-bit integer global seed, and let `node_path` be the string representation of the path from the root to node v (e.g., "0/1/0" for the left-right-left child of the root). The CVD node seed is*

$$s_v = \text{first8}\left(\text{SHA-256}(\text{"CVD_INIT_"} \parallel \text{node_path} \parallel \text{"_"} \parallel \text{le8}(\text{base_seed}))\right),$$

where $\text{le8}(x)$ encodes x as 8 little-endian bytes and $\text{first8}(h)$ reads the first 8 bytes of hash h as a little-endian unsigned 64-bit integer [National Institute of Standards and Technology, 2015].

The domain-separation prefix "CVD_INIT_" ensures that the CVD seed is independent

from all other SHA-256 applications in the platform (e.g., "SA_NODE_" for U.3's simulated annealing). A platform invariant test asserts this prefix constant equals "CVD_INIT_" exactly, and that $s_v(\text{base_seed} = 0)$ equals a hard-coded expected value.

3.3. K-Farthest Seed Initialisation

The two initial seeds are placed by the k-farthest BFS procedure:

Definition 2 (K-Farthest Initialisation). *Let the tracts in V_v be sorted by global index, giving an ordered list $\text{sorted} = (t_0, t_1, \dots, t_{m-1})$.*

1. **First seed:** $\text{seed}_0 = t_{s_v \bmod m}$, where s_v is the CVD node seed.
2. **BFS from seed_0 :** compute $d_0(t) = d_{\text{BFS}}(\text{seed}_0, t)$ for all $t \in V_v$.
3. **Second seed:** $\text{seed}_1 = \arg \max_{t \in V_v} d_0(t)$ (farthest tract from seed_0 ; ties broken by sorted index).

K-farthest initialisation maximises the initial distance between the two seeds, preventing the degenerate configuration where both seeds begin in the same geographic region. For path-like subgraphs (rural county chains), k-farthest trivially places seeds at the two endpoints of the path; Voronoi assignment then bisects the path at the midpoint, and convergence occurs in 1–2 iterations. For complex urban subgraphs with multiple geographic centres, the farthest-pair seeds maximise the chance that each seed captures a distinct region.

Remark 1. *The sorted-index tie-breaking in both the first seed selection ($s_v \bmod m$, which maps the hash to a position in the sorted array) and the $\arg \max$ ensures that the same `base_seed` and `node_path` always produce the same seed pair, regardless of the order in which tracts are processed by the surrounding parallelism. This reproducibility invariant is tested at L0.*

3.4. Voronoi Assignment and Medoid Update

Definition 3 (Graph Voronoi Assignment). *Given seeds $(\text{seed}_0, \text{seed}_1)$, the Voronoi assignment $a : V_v \rightarrow \{0, 1\}$ assigns each tract to its nearest seed by BFS distance:*

$$a(t) = \arg \min_{j \in \{0, 1\}} d_{\text{BFS}}(t, \text{seed}_j),$$

with ties broken in favour of district 0.

Multi-source BFS computes both $d_{\text{BFS}}(\cdot, \text{seed}_0)$ and $d_{\text{BFS}}(\cdot, \text{seed}_1)$ simultaneously in a single BFS pass seeded from both seeds, costing $O(m)$ per assignment step.

Definition 4 (Approximate Graph-Distance Medoid). *Let $D_j = \{t \in V_v : a(t) = j\}$ be the tracts assigned to district j . Let $\text{probe}_j \subseteq D_j$ be a uniform random sample of at most $n_{\text{probe}} = 50$ tracts from D_j (or all tracts if $|D_j| \leq 50$). The approximate medoid of D_j is*

$$\text{medoid}_j = \arg \min_{i \in \text{probe}_j} \sum_{t \in D_j} d_{\text{BFS}}(i, t).$$

The probe set limits the number of BFS evaluations per district per iteration to at most 50, making each medoid update $O(50 \times m)$ in the worst case. For $|D_j| \leq 50$, the probe set equals D_j and the medoid is exact.

3.5. Formal Pseudocode

Algorithm 1 gives the complete CVD bisection procedure.

3.6. Three Key Design Decisions

Decision 1: K-farthest seed placement. Naïve seed initialisation (e.g., placing both seeds at the first two tracts in sorted order) would typically start both seeds in the same geographic corner of the subgraph, requiring many iterations to separate. K-farthest initialisation places seeds at the two most geographically distant tracts in the subgraph, maximising coverage from the first iteration. This reduces the typical iteration count from ~ 20 (random initialisation) to ~ 10 (k-farthest) on NC-scale subgraphs.

Decision 2: Medoid, not centroid. In continuous Euclidean space, the centroid of a cluster is the well-defined population-weighted mean position. In discrete graph space, the mean of a set of tract indices is a fraction — a position that may not correspond to any real tract, and that cannot be used as a BFS source in subsequent Voronoi assignments. The graph-distance medoid avoids this problem by requiring the new seed to be an actual tract in its district. Medoids are also more robust to outliers: the medoid minimises absolute distance (L1 sense in graph space), whereas the centroid minimises squared distance (L2 sense).

Algorithm 1 CVD-BISECT(G_v , k_L , k_R , base_seed , node_path , cvd_iters , balance_tolerance)

```

1:  $m \leftarrow |V_v|$ 
2: if  $m < 2$  then return degenerate assignment ▷ fallback:  $\leq 1$  tract
3: end if
4:  $s_v \leftarrow \text{CVD-Node-Seed}(\text{base\_seed}, \text{node\_path})$  ▷ Definition 1
5:  $\text{sorted} \leftarrow \text{sort}(V_v)$  ▷ deterministic ordering
6:  $\text{seed}_0 \leftarrow \text{sorted}[s_v \bmod m]$ 
7:  $d_0 \leftarrow \text{BFS-Distances}(\text{seed}_0, G_v)$ 
8:  $\text{seed}_1 \leftarrow \arg \max_{t \in V_v} d_0[t]$  ▷ k-farthest: Definition 2
9:  $\text{seeds} \leftarrow (\text{seed}_0, \text{seed}_1)$ 
10:  $\text{prev\_seeds} \leftarrow (-1, -1)$  ▷ sentinel: no previous seeds
11: for  $\text{iter} = 0$  to  $\text{cvd\_iters} - 1$  do
12:    $a \leftarrow \text{VoronoiAssign}(\text{seeds}, G_v)$  ▷ multi-source BFS: Definition 3
13:   for  $j \in \{0, 1\}$  do
14:      $D_j \leftarrow \{t : a(t) = j\}$ 
15:      $\text{probe}_j \leftarrow \text{Sample}(D_j, \min(50, |D_j|))$ 
16:      $\text{seeds}[j] \leftarrow \arg \min_{i \in \text{probe}_j} \sum_{t \in D_j} d_{\text{BFS}}(i, t)$  ▷ approx. medoid: Definition 4
17:   end for
18:   if  $\text{seeds} = \text{prev\_seeds}$  then break ▷ convergence: seeds unchanged
19:   end if
20:    $\text{prev\_seeds} \leftarrow \text{seeds}$ 
21: end for
22:  $a \leftarrow \text{VoronoiAssign}(\text{seeds}, G_v)$  ▷ final assignment from converged seeds
23:  $(S, \bar{S}) \leftarrow \text{Rebalance}(a, G_v, \mathbf{p}, \text{balance\_tolerance})$  ▷ 200-iter boundary-swap
24: return  $(S, \bar{S})$ 

```

Decision 3: Post-hoc rebalance. Voronoi assignment by BFS distance does not guarantee population balance. In a state with highly unequal tract populations, the geometrically central tract may be much more or less populous than others at equal hop count. The 200-iteration boundary-swap rebalance (consistent with the existing `run_multiscale` rebalance) corrects population imbalance by moving boundary tracts from the overweight district to the underweight district, one tract at a time, while preserving contiguity. Without this step, CVD plans would frequently violate the federal $\pm 0.5\%$ population balance tolerance.

3.7. Convergence Analysis

Proposition 1 (Monotone Energy Descent). *Let $\mathcal{E}(\text{seeds}) = \sum_j \sum_{t \in D_j} d_{\text{BFS}}(t, \text{seeds}[j])$ be the total graph-distance quantisation energy. Each complete iteration (Voronoi assignment + exact medoid update) satisfies $\mathcal{E}(\text{seeds}^{(i+1)}) \leq \mathcal{E}(\text{seeds}^{(i)})$.*

Proof. Two observations: (a) Voronoi assignment minimises $\mathcal{E}$ over all assignments given fixed seeds: for each tract t , assigning to the nearer seed (by d_{BFS}) reduces or maintains $\sum_t d_{\text{BFS}}(t, \text{seeds}[a(t)])$. (b) Medoid update minimises $\mathcal{E}$ over all seed choices given fixed assignment: by definition, $\text{medoid}(D_j)$ minimises $\sum_{t \in D_j} d_{\text{BFS}}(i, t)$ over $i \in D_j$. Combining (a) and (b): each full iteration is a coordinate descent step on $\mathcal{E}$ (alternating over assignments and seed positions), which is non-increasing. Since $\mathcal{E} \geq 0$ and takes values in a finite set (there are finitely many tract pairs), the sequence $\mathcal{E}(\text{seeds}^{(i)})$ converges in finitely many steps. $\square$

Remark 2. *Proposition 1 applies to the exact medoid. The approximate medoid (probe size 50) does not guarantee monotone descent. In practice, the approximation error is small: for large districts ($|D_j| \gg 50$), the probe median concentrates near the true medoid, and the energy is nearly monotone. For small districts ($|D_j| \leq 50$), the probe is exact and the guarantee holds.*

Remark 3 (Approximate medoid error bound). *An approximate medoid with probe size p has expected relative error bounded by $O(1/\sqrt{p})$ for the objective function value (the sum of BFS distances). For $p = 50$: the probability that the true medoid lies outside the sampled set is $\leq (m - p)/m \leq 1 - 50/m$. In practice on NC bisection nodes ($m \approx 160$ tracts), $p = 50$ covers 31% of the district, giving expected convergence within 3–4 extra iterations at worst. We empirically observe that convergence is unaffected by probe size for $p \geq 30$ on all test states.*

Remark 4 (Path-like graphs.). *For subgraphs that are paths (a sequence of tracts $t_0 - t_1 - \dots - t_{m-1}$), k -farthest initialisation places seed_0 and seed_1 at the two endpoints. The Voronoi assignment bisects the path at the midpoint. The medoid of each half is the midpoint of that half. After one iteration, the seeds are at the midpoints of each half, and the Voronoi assignment is unchanged. Convergence occurs at iteration 2.*

3.8. Computational Complexity

Proposition 2 (Runtime Complexity). *CVD bisection at a node with m tracts requires:*

- *One BFS for k -farthest initialisation: $O(m)$.*
- *Per iteration: two multi-source BFS passes ($O(m)$ each) for Voronoi assignment, plus at most 2×50 single-source BFS evaluations ($O(m)$ each) for medoid update.*

- *Total per iteration:* $O(52 \times m) = O(m)$.
- *Over n_{iter} iterations:* $O(n_{\text{iter}} \times m)$.
- *Post-hoc rebalance:* $O(200 \times m)$ for boundary-swap iterations.

Total: $O((n_{\text{iter}} + 200) \times m)$. For the default `cvd_iters = 20`, this is $O(220 \times m) = O(m)$ per bisection node.

Comparing to METIS ($O(m \log m)$ per node) and SIMULATEDANNEALING ($O(\text{steps} \times m^{3/2})$ per node in the planar-graph regime): CVD is $O(m)$ per node, asymptotically faster than both on large subgraphs. In practice, CVD and METIS have similar wall-clock times because the BFS constant in CVD is larger than the multilevel coarsening constant in METIS.

3.9. CLI Invocation

CVD bisection is exposed via `-structure centroidal-voronoi` in the `bisect` binary:

```
bisect state --state NC --year 2020 \
  --structure centroidal-voronoi --cvd-iters 20
```

With explicit metric and seeding options (Phase 2 preview):

```
bisect state --state NC --year 2020 \
  --structure centroidal-voronoi --cvd-iters 20 \
  --cvd-metric graph-distance --cvd-seed-init kmeans++
```

The YAML configuration format:

```
algorithm:
  structure: centroidal-voronoi
  weights: geographic
  search: convergence
  cvd_iters: 20
  cvd_metric: graph-distance
  cvd_seed_init: kmeans++
  convergence_threshold: 600
  balance_tolerance: 0.5
workers: 8
```

3.10. Audit Chain

Every CVD run appends structured metadata to `runs/{label}/{year}/index.json`:

```
"structure": "centroidal-voronoi",
"n_iter": 20,
"metric": "graph-distance",
"seed_init": "kmeans++",
"base_seed": 12345678,
"cvd_seed": 987654321,
"iterations_to_convergence": 14,
"convergence_warning": false,
"final_ec": 2847,
"final_pp_mean": 0.241
```

`iterations_to_convergence` records how many iterations were needed before seeds stabilised. If `iterations_to_convergence == n_iter`, the algorithm did not converge within the budget; `convergence_warning` is set to `true` as a non-fatal diagnostic. An auditor who knows `base_seed` can recompute `cvd_seed = SHA-256("CVD_INIT_" || base_seed)` and verify full reproducibility of the plan.

4. Comparison with METIS and Simulated Annealing

4.1. Experimental Setup

We compare three algorithms on three states that span the range of congressional delegation sizes and geographic complexity:

- **NC** (North Carolina, $k = 14$, $n \approx 2,660$ tracts): medium complexity, competitive two-party state with a mix of urban, suburban, and rural geography.
- **WI** (Wisconsin, $k = 8$, $n \approx 1,900$ tracts): smaller delegation, historically contested, elongated east-west shape.
- **TX** (Texas, $k = 38$, $n \approx 5,300$ tracts): large delegation, geographically diverse with major metropolitan areas and vast rural west.

All runs use 2020 Census data, geographic boundary-length edge weights (the B.2 default [Della-Libera, 2026b]), the standard bisection tree structure, and a single seed ($s = s_0$, the content-derived seed). The three algorithms compared are:

1. **METIS (baseline)**: Single METIS call per node, no refinement. This is the standard platform configuration [Karypis and Kumar, 1998].
2. **SA(steps=10)**: SIMULATEDANNEALING bisection with `steps_per_tract = 10`, warm-started from the METIS solution [Della-Libera, 2026a].
3. **CVD(iters=20)**: CVD bisection with `cvd_iters = 20`, k-farthest seed initialisation, approximate medoid (probe size 50), and 200-iteration rebalance.

Metrics. For each (state, algorithm) pair we report:

- **Edge Cut (EC)**: Total weighted edge cut of the final k -district plan; lower is better.
- **Polsby-Popper (PP)**: Mean PP compactness across all k districts; higher is more compact. Baseline values from U.9 [Della-Libera, 2026e]: NC PP = 0.217, WI PP = 0.198, TX PP = 0.184.
- **Iterations**: For CVD, the number of iterations to convergence.
- **Runtime (s)**: Wall-clock time on a single core, 3.5 GHz CPU.

4.2. Results

Table 2 presents the comparison. CVD(iters=20) achieves Polsby-Popper compactness competitive with METIS — within 5% on NC and WI — despite not optimising edge cuts. The edge-cut values for CVD are substantially higher than METIS, as expected: CVD does not minimise edge cuts, and the Voronoi assignment by BFS distance is not designed to follow geographic boundary lines.

4.3. Analysis of Results

Edge cut. CVD produces substantially higher edge cuts than METIS across all three states (NC: +17%, WI: +20%, TX: +15%). This is expected: CVD assigns tracts by geographic

Table 2: Algorithm comparison on NC ($k = 14$), WI ($k = 8$), TX ($k = 38$). EC: total edge cut (lower is better). PP: mean Polsby-Popper compactness (higher is better). Iter: iterations to convergence (CVD only; “—” for others). Runtime: wall-clock seconds (single core). Δ PP: PP relative to METIS baseline (percentage points).

State	Algorithm	EC	PP	Δ PP (%)	Iter	Time (s)
NC ($k = 14$)	METIS (baseline)	4,821	0.217	—	—	4.2
	SA (steps=10)	4,513	0.248	+14.3	—	51.7
	CVD (iters=20)	5,639	0.231	+6.5	13	6.8
WI ($k = 8$)	METIS (baseline)	2,103	0.198	—	—	1.8
	SA (steps=10)	1,962	0.220	+11.1	—	20.3
	CVD (iters=20)	2,517	0.206	+4.0	11	2.9
TX ($k = 38$)	METIS (baseline)	11,440	0.184	—	—	18.3
	SA (steps=10)	10,729	0.204	+10.9	—	227
	CVD (iters=20)	13,208	0.189	+2.7	15	24.6

proximity, not by following graph boundaries. Voronoi boundaries frequently cross many adjacency edges because the bisection boundary is a straight path through the graph (a geodesic in hop-count space) rather than a minimum-cut boundary.

Polsby-Popper compactness. Despite higher edge cuts, CVD achieves PP compactness *above* the METIS baseline on all three states. The mechanism is geometric: Voronoi regions centred at well-separated seeds tend to be roughly circular (or at least convex in graph space), which correlates with geographic compactness. METIS minimises edge cuts but produces plans shaped by the graph’s connectivity structure, which may follow non-compact county or road boundaries. CVD’s PP improvement over METIS is modest (2.7–6.5% depending on state), well below SA’s 10.9–14.3%, but CVD achieves it without any edge-cut-based objective.

Compactness ordering. The observed quality ordering is:

$$\text{METIS} < \text{CVD} < \text{SA}(\text{steps}=10),$$

where $<$ denotes lower PP (worse compactness). This ordering reflects the different compactness mechanisms: METIS minimises boundary length (edge cut) but has no explicit compactness target; CVD targets geometric proximity but not boundary minimisation; SA refines METIS by escaping local optima in the edge-cut landscape.

Convergence. CVD converges in 11–15 iterations across the three states, well within the default `cvd_iters` = 20 budget. NC converges in 13 iterations; WI (more elongated) in 11; TX (more complex urban geography) in 15. No convergence warnings were triggered. Convergence is faster on elongated states (WI) because the k-farthest initialisation places seeds near the eastern and western ends of the state, requiring fewer iterations to stabilise.

Runtime. CVD(iters=20) is approximately $1.6\times$ slower than METIS on NC (6.8 s vs. 4.2 s) and approximately $7.6\times$ faster than SA(steps=10) (6.8 s vs. 51.7 s). The runtime ratio CVD/METIS is approximately constant across states (NC: $1.6\times$, WI: $1.6\times$, TX: $1.3\times$), consistent with the $O(n_{\text{iter}} \times m)$ scaling predicted by Proposition 2 with $n_{\text{iter}} \approx 13$ iterations.

Partisan outcomes. Table 2 omits partisan seat counts because CVD produces geometrically different maps from METIS: the seat distributions vary at the ± 1 seat level on NC and WI, and at the ± 2 seat level on TX. This variability reflects the different geographic structure of CVD plans versus METIS plans, and is consistent with the B.7 finding that partisan outcomes are sensitive to structure-layer choices. A full 50-state partisan analysis of CVD is deferred to G.14 [Della-Libera, 2026d].

Table 3 reports partisan seat counts (2020 presidential two-party returns) for CVD plans compared to standard bisection.

State	k	CVD	Standard METIS	Single-run note
NC	14	7D/7R	7D/7R	Stable
WI	8	3D/5R [†]	2D/6R [†]	One-seat variation
TX	38	16D/22R	16D/22R	Stable

Table 3: Partisan outcomes (2020 presidential, single run, seed $s = 42$). CVD produces proportional outcomes consistent with ApportionRegions, consistent with the geographic-proximity mechanism being partisan-neutral.

[†] Single-run result (seed=42); partisan seat counts are seed-sensitive for competitive states. Multi-seed validation (50 runs) deferred to Phase 2.

4.4. The CVD Compactness Profile

The Polsby-Popper results above suggest that CVD occupies a distinct position in the compactness-vs-boundary-cost trade-off space. METIS minimises boundary cost (edge cut) and achieves modest compactness as a side effect. SIMULATEDANNEALING further minimises boundary cost through annealing and achieves higher compactness. CVD ignores boundary cost entirely and achieves compactness through a completely different mechanism: geographic proximity.

This orthogonality is potentially useful in legal contexts. A plaintiff challenging a map as an extreme outlier needs to characterise the space of “normal” maps. A space containing only edge-cut-minimising plans (METIS, SA) may not capture the full range of geometrically reasonable alternatives. Adding CVD plans to the ensemble ensures that the “natural partition” criterion is represented, not just the “boundary-minimising partition” criterion.

PP distribution overlap. To assess whether CVD plans inhabit a geometrically distinct region of plan space (rather than merely offset METIS plans), we computed the PP histogram overlap between 100 METIS seeds and 20 CVD variants (different base seeds on NC $k = 14$). The CVD PP distribution is centred at $\mu_{\text{CVD}} = 0.242 \pm 0.008$ and the METIS PP distribution at $\mu_{\text{METIS}} = 0.217 \pm 0.011$; the distributions have $< 15\%$ overlap, confirming that CVD plans occupy a geometrically distinct region of plan space.

4.5. Relation to BisectionEnsemble

CVD is a Structure-layer algorithm; BISECTIONENSEMBLE [Della-Libera, 2026e] is a SeedCompositor-layer strategy. Running BISECTIONENSEMBLE($T = 50$) with CVD bisectioners would execute 50 independent CVD runs per node, each with a different base seed (and therefore a different CVD node seed and different k-farthest initialisation), and select the result with the lowest edge cut. This combination is expected to improve CVD’s PP further, since the worst-EC CVD plans tend to also be less compact. This configuration is left for G.14 [Della-Libera, 2026d].

5. Phase 2: Geographic Voronoi Districts

5.1. Motivation: When Graph Distance Diverges from Geography

Phase 1 CVD measures proximity via BFS hop count on the census-tract adjacency graph. BFS hop count is a faithful proxy for geographic distance in most of the contiguous United States: inland states with roughly regular census-tract shapes produce adjacency graphs where hop count is proportional to road distance, and Voronoi regions based on hop count are reasonable geographic approximations.

However, BFS hop count diverges from geographic Euclidean distance in two important cases:

Case 1: Coastal and waterfront states. States with substantial water boundaries (Florida, Louisiana, Michigan, Maryland) have census tracts that are geographically far apart but connected via many intermediate tracts along land corridors. Two tracts on opposite shores of a bay may be 2 miles apart in Euclidean distance but 50 hops apart in the adjacency graph (via the land route). BFS Voronoi in such states assigns coastal tracts to “wrong” seeds based on graph topology rather than actual geographic proximity, producing elongated peninsular districts rather than compact coastal ones.

Case 2: Elongated rural states. States with elongated geographic shapes and irregular census-tract sizes (Nevada, Montana, Wyoming) have rural tracts that are large in area but contribute only 1 hop to BFS distance. A rural tract covering 1,000 square miles is treated the same as an urban tract covering 1 square mile in hop-count space. BFS Voronoi places district boundaries in locations that are geometrically natural in hop-count space but geographically arbitrary.

Phase 2 CVD addresses both cases by replacing BFS hop count with Euclidean distance on projected tract centroids.

5.2. Geographic CVD: Algorithm Modification

The Phase 2 modification to Algorithm 1 is minimal:

1. **Data requirement:** Load a `tract_centroids` field in `LoadedGraph`: `pub tract_centroids: Vec<(f64, f64)>`, containing the (longitude, latitude) of each tract’s polygon centroid.

Centroids are available from TIGER/Line shapefiles as the centroid of each tract polygon, derivable from the `_geoids.json` file already present in the pipeline.

2. **Metric change:** Replace $d_{\text{BFS}}(t, \text{seed}_j)$ with $d_{\text{Euc}}(t, \text{seed}_j)$, the Euclidean distance between projected centroids.
3. **Centroid update:** Replace the graph-distance medoid with the population-weighted centroid:

$$\text{centre}_j = \frac{\sum_{t \in D_j} p_t \cdot \mathbf{x}_t}{\sum_{t \in D_j} p_t},$$

where $\mathbf{x}_t = (x_t, y_t)$ is the projected centroid of tract t . This is the true Euclidean centroid, not a medoid. The nearest tract to the centroid (by Euclidean distance) is used as the new seed for subsequent Voronoi assignments.

4. **Projection:** Use EPSG:5070 Albers Equal Area Conic for all 50 states [EPSG Geodetic Parameter Dataset, 2023]. For continental US states, raw (lon, lat) Euclidean distance is adequate; for Alaska and Hawaii, where meridian convergence is severe, EPSG:5070 is mandatory to avoid distortion. Using EPSG:5070 uniformly ensures consistency with the projection used in `redist-map` for area and compactness computations.

5.3. Expected Improvement over Phase 1

Phase 2 geographic CVD is expected to yield a 5–10% Polsby-Popper improvement over Phase 1 graph-distance CVD for states where BFS hop count diverges substantially from geographic distance.

The intuition: Euclidean Voronoi regions on a geographic plane are exactly the geometric cells that minimise mean squared distance from seed to tract centroid. These cells are circular in uniform-density regions, and elliptical in anisotropic regions, both of which are more compact (higher PP) than the irregular BFS Voronoi regions. For states where hop count faithfully proxies geography (Iowa, Kansas), the Phase 2 improvement is expected to be small ($< 3\%$). For coastal states (Florida, Louisiana), the improvement is expected to be larger (5–10%) because Phase 2 avoids the along-shore bias of hop-count Voronoi.

5.4. Implementation Plan

Phase 2 CVD is not yet implemented; it is deferred pending the addition of `tract_centroids` to `LoadedGraph`. The implementation plan is:

1. **Add `tract_centroids` to `LoadedGraph`:** Derive centroid (lon, lat) for each tract from the existing TIGER/Line shapefile adjacency loader. Store as a `Vec<(f64, f64)>` field in the `LoadedGraph` struct. Add to the `.pkl` adjacency cache for zero-cost access on subsequent runs.
2. **Implement `VoronoiMetric::Geographic`:** Add the Euclidean distance branch to the CVD Rust implementation. Project coordinates to EPSG:5070 before computing distances using the existing `redist-map` projection utilities.
3. **Validate on FL and LA:** Florida and Louisiana are the strongest tests because of their complex coastal geometries. Compare Phase 1 and Phase 2 CVD PP on these states to confirm the expected improvement.
4. **Expose via CLI:** `-cvd-metric geographic` selects Phase 2. The default remains `graph-distance` for backward compatibility.

5.5. Connection to CVT Theory

Phase 2 geographic CVD is directly the continuous CVT construction of Du et al. 1999 applied to the discrete redistricting domain. With Euclidean distance, the Phase 2 algorithm satisfies the two CVT conditions: (1) each tract is assigned to its nearest seed (Voronoi condition); (2) each seed is at the population-weighted centroid of its district (centroidal condition, up to the discretisation of approximating a continuous centroid by the nearest actual tract).

The convergence guarantee of Proposition 1 applies to Phase 2 without modification: Euclidean Voronoi assignment minimises the sum of squared distances given fixed seeds, and centroid update minimises the same sum given fixed assignment. The discrete rounding (nearest tract to the continuous centroid) introduces a rounding error, but does not break the monotone energy descent as long as the error is smaller than the step size.

The convergence rate for Phase 2 is expected to be faster than Phase 1 in Euclidean-like geographies: the Lloyd energy surface for Euclidean distance is smoother and has fewer local

minima than the graph-distance surface, because Euclidean distance is a continuous metric while BFS hop count is discrete and highly non-convex.

6. Conclusion

We have introduced Centroidal Voronoi Districts (CVD), a Structure-layer algorithm that assigns census tracts to district seeds by graph BFS distance and iterates seeds to the population-weighted graph-distance medoid of their assigned districts. CVD is the discrete redistricting analogue of Lloyd’s algorithm for optimal quantisation [Lloyd, 1982, Du et al., 1999], adapted to weighted planar graphs via the graph-distance medoid as the centroid surrogate.

Position in the B-series algorithm landscape. CVD is a geometric structure algorithm, complementing the graph-topology structure algorithms (METIS, SIMULATEDANNEALING, GEOSECTION, AREASECTION). The two families serve different purposes:

- **Graph-topology methods** (METIS, SIMULATEDANNEALING): minimise boundary length (edge cut), producing plans that follow roads, rivers, and county boundaries. Best for: publication-quality single-plan generation when compactness is measured by boundary minimisation.
- **Geometric methods** (CVD, BFSGROWTH): assign by proximity, producing plans that fill geographic space radially. Best for: geometric baseline plans, compactness diversity in ensemble analysis, and “most natural partition” legal arguments.

The empirical compactness ordering observed in Section 4 is:

$$\text{METIS} < \text{CVD}(\text{iters}=20) < \text{SA}(\text{steps}=10),$$

where $<$ denotes lower PP (worse compactness). CVD sits between METIS and SIMULATEDANNEALING in compactness, at approximately METIS runtime ($1.6\times$ slower on NC vs. $12\times$ for SIMULATEDANNEALING).

When to use CVD.

- **Geometric baseline:** CVD provides the “most geographically natural partition” — the Voronoi tessellation of census tracts around their nearest district seed. This is a legally defensible baseline for redistricting challenges.
- **Ensemble diversity:** Adding CVD plans to a redistricting ensemble enriches the distribution with geometrically compact plans that are structurally different from edge-cut-minimising plans.
- **Fast initialisation:** At $O(n_{\text{iter}} \times m)$ runtime, CVD is the fastest compactness-improving structure method in the B-series. For exploratory runs, CVD provides a valid, balanced, compact plan at a cost close to METIS.

Algorithmic baselines in redistricting litigation. Voronoi-based and geometric partitioning algorithms have been proposed as objective baselines in redistricting research (see Herschlag et al. 2020; Chen & Rodden 2013 on geographic baselines). To our knowledge, no published redistricting court opinion has specifically cited a Voronoi partition as evidence; practitioners should frame CVD as a *compactness reference*, not as a legally binding standard. The value of CVD for litigation is comparative: showing that a contested plan performs worse than the simplest geometric baseline on compactness measures.

Phase 2 geographic CVD. Section 5 previewed Phase 2 CVD using Euclidean distance on EPSG:5070 projected tract centroids. Phase 2 is expected to yield a 5–10% further PP improvement for coastal and elongated states where BFS hop count diverges from geographic distance. Implementation requires adding `tract_centroids` to `LoadedGraph`, which is a Phase 2 data pipeline task.

Implementation. CVD bisection is implemented in the `bisect` binary under `-structure centroidal-voronoi`. The primary tunable parameter is `-cvd-iters` (default 20). The YAML configuration format:

```
algorithm:
  structure: centroidal-voronoi
  cvd_iters: 20
  cvd_metric: graph-distance
  balance_tolerance: 0.5
```

Open questions. Several extensions are identified for future work:

1. **Phase 2 geographic CVD** (Section 5): implement Euclidean Voronoi once `tract_centroids` is added to `LoadedGraph`. Validate on Florida and Louisiana for the largest expected improvements.
2. **CVD + BisectionEnsemble**: run `BISECTIONENSEMBLE( $T = 50$ )` with CVD bisectors to combine geometric proximity with multi-seed selection. This is expected to improve PP over single-seed CVD by selecting the most compact geometric Voronoi plan across 50 initialisations.
3. **CVD as SA warm start**: use the CVD plan as the initialisation for `SIMULATEDANNEALING` bisection instead of the METIS plan. Since CVD provides a geometrically compact starting point, `SIMULATEDANNEALING` refinement from CVD may find lower-EC solutions with different compactness properties than `SIMULATEDANNEALING` refinement from METIS.
4. **$k > 2$ CVD**: The current implementation bisects ($k = 2$) at each node. A direct k -way CVD that places k seeds simultaneously is a natural extension, applicable at nodes where the bisection tree strategy calls for a k -way split (as in the `nway` structure or prime-factor nodes with $k_v = p$ for odd primes p).
5. **Convergence for approximate medoid**: Formalise the convergence analysis for the approximate medoid (probe size 50). Empirical evidence suggests monotone descent in practice; a formal bound on the approximation error’s effect on convergence rate would complete the theoretical picture.

These extensions inherit the clean three-layer compositor design and require no structural changes to the codebase beyond the Phase 2 data field addition. CVD provides a principled geometric complement to the graph-topology structure methods in the B-series, rounding out the algorithm portfolio with a method that fills geographic space by proximity rather than by boundary minimisation.

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